

TRUNG NGUYEN

413-425-3931 | trungnguyenquochn2202@gmail.com | [linkedin.com/in/trung-nguyen-hn2202](https://www.linkedin.com/in/trung-nguyen-hn2202) | github.com/Trungsherlock

EXPERIENCE

FPT Software – A Global IT Services Company

Jun 2022 – Sep 2022

Software Engineer Intern

Hanoi, Vietnam

- Developed internal interview management system using Node.js, Express.js, and MySQL on AWS EC2, scaling across **4 business units for 200+ recruiters**, reducing interview setup from 60 to 20 minutes (**67% reduction**) and eliminating **3 hours of weekly manual coordination**.
- Implemented an automated question bank, generating role-specific assessments in **under 30 seconds** compared to 15-minute manual creation, while maintaining a **90% relevance score** based on interviewer feedback.

MLSys Lab – University of Massachusetts, Amherst

Jun 2023 – Sep 2023

Machine Learning Researcher

Amherst, MA

- Optimized BERT model for mobile deployment at scale, **achieving 30% memory reduction** (from 420MB to 294MB) through quantization and pruning techniques while maintaining 97.5% of original accuracy for production ML.
- Developed custom PyTorch optimization pipeline for LoRA fine-tuning that **reduced training time by 45%** and enabled efficient deployment on edge devices with **<500ms inference latency**.

OPEN SOURCE CONTRIBUTIONS

CYOBot LLC – A Robotics-based CS Education Startup

May 2024 – Sep 2025

- Built containerized robotics education platform using Next.js, and PayloadCMS **serving 10+ educational institutions with 50+ monthly active users**.
- Architected backend infrastructure with PostgreSQL optimization and integrated multiple APIs (GitHub, Shopify, Discord), **reducing API response time by 40%** and enabling proactive error detection through Sentry monitoring that decreased bug resolution time from 48 to 12 hours.

PROJECTS

FoundIt - Project Discovery Platform | Next.js, Prisma, Supabase, Redis, PostgreSQL, OAuth

- Built high-performance API with **Redis caching achieving 10-40x query acceleration** and PostgreSQL indexing; optimized database schema with connection pooling to deliver **sub-100ms response times at 3,000+ entity scale**.
- Developed production infrastructure with OAuth 2.0 multi-provider authentication, Redis-based rate limiting, and **complete monitoring stack** (Prometheus, Grafana, health checks) supporting horizontal scalability.

Datahotpot - Web3 Data Exchange | Next.js, IPFS, Web3 APIs, Docker, AWS S3

- Built decentralized data marketplace with NFT-based ownership verification; earned **top-5 placement** in FVM Space Warp Hackathon.
- Integrated IPFS storage with S3 backup strategy and **Redis caching, reducing upload latency by 30%** and ensuring data reliability across distributed nodes.

Echo - AI Customer Support Platform | Next.js, React, Convex, AWS Secrets Manager, Sentry

- Built AI customer support platform using Next.js, React, and Convex Agents with real-time chat, voice support via VAPI, and RAG-powered knowledge base serving B2B SaaS customers.
- Implemented secure cloud infrastructure with AWS Secrets Manager for API key management, IAM policies with tenant-specific access control, and Sentry monitoring integrated with an automated CI/CD pipeline.

EDUCATION

University of Massachusetts, Amherst

Amherst, MA

Master of Science in Computer Science

Expected Graduation: Dec 2025

Bachelor of Science in Computer Science and Mathematics

GPA: 3.9

TECHNICAL SKILLS

Languages & Frameworks: TypeScript, JavaScript, Python, SQL, Next.js, React.js, Node.js, Express.js

Backend: Prisma ORM, PostgreSQL, Redis, RESTful APIs, OAuth 2.0, NextAuth.js, Rate Limiting, Query Optimization

DevOps & Infrastructure: Docker, GitHub Actions, Prometheus, Grafana, Jenkins, CI/CD

Cloud & Tools: AWS (EC2, S3, Lambda), MySQL, MongoDB, Firebase, Sentry, Linux

CERTIFICATIONS

AWS Certified Cloud Practitioner

Oct 2025

Amazon Web Services